

UNIT II

1. AC CIRCUITS

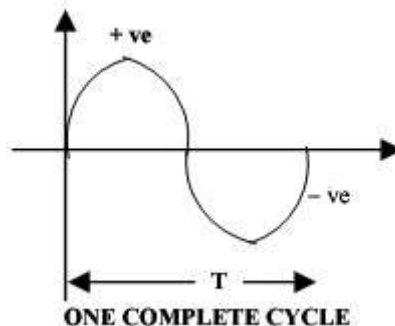
2.1 TERMINOLOGIES IN AC CIRCUITS

2.1.1 Alternating Quantity:

An alternating quantity changes its magnitude and direction periodically at regular intervals.

2.1.2 Cycle:

One complete set of positive and negative values of an alternating quantity constitutes one cycle.



2.1.3 Time Period:

It is the time taken by an alternating quantity to complete one cycle.

E.g. 50 Hz alternating quantity has a time period of $\frac{1}{50}$ sec.

(i.e.) $T = \frac{1}{f}$

Where, T – Time period

f – Supply frequency

2.1.4 Frequency:

The number of cycles completed per second is known as frequency. Its unit is cycles/sec (or) Hertz (Hz) and it is denoted “ f ”

$$f = \frac{1}{T}$$

2.1.5 Amplitude:

The maximum value (+ve or –ve) of an alternating quantity is called amplitude.

2.1.6 RMS Value (Root Mean Square value or Effective Value)

The RMS value of alternating current is given by the value of the direct current which when flowing through a given circuit for a given time produces the same heating effect as that produced by an alternating current when flowing through the same circuit for the same time.

It is also called virtual value or effective value.

$$RMS \text{ Value} = \sqrt{\frac{\text{Area under the wave}}{\text{period}}}$$

$$I_{RMS} = \sqrt{\frac{1}{2\pi} \int_0^{2\pi} i^2 dt}$$

$$V_{RMS} = \sqrt{\frac{1}{2\pi} \int_0^{2\pi} v^2 dt}$$

2.1.7 Average value or Mean value

It is the average of the instantaneous value taken over one complete cycle of the wave.

Average value can be obtained by finding the area under the curve and dividing it by base period.

$$\text{Average Value} = \frac{\text{Area under complete cycle}}{\text{Base period}}$$

2.1.8 Form factor (K_F)

Form factor is the ratio of RMS value to average value.

$$(K_F) = \frac{RMS \text{ Value}}{Average \text{ Value}}$$

2.1.9 Peak/crest/amplitude factor

Peak factor is the ratio of maximum value to RMS value.

$$(K_p) = \frac{Maximum \text{ Value}}{RMS \text{ Value}}$$

2.1.10 Power factor

Power factor is the ratio of active power to apparent power. It is given by “ $\cos \phi$ ” where ϕ is the phase angle between voltage and current.

$$P.F(\cos \phi) = \frac{Active \text{ Power}}{Apparent \text{ Power}}$$

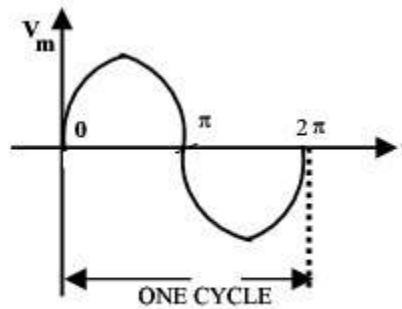
2.1.11 Phase angle

When two alternating quantities do not reach their zero or maximum value at the same time, they are said to be out of phase by some angle. This is known as phase angle and it is measured in degree (or) radians.

It can be leading phase angle or lagging phase angle.

Problems

2.1 Calculate average value, RMS value, Peak factor and Form factor for the given Sinusoidal waveform.



$$V = V_m \sin \theta$$

$$\text{Where } \theta = \omega t$$

Since it is a symmetrical wave, consider one half of cycle only.

$$\omega = 2\pi f$$

$$f = \frac{1}{T}$$

$$\text{Average value} = \frac{\text{area under one cycle}}{\text{Base (or) time period}}$$

$$\text{Average value} = \frac{1}{\pi} \int_0^{\pi} V dt$$

$$= \frac{1}{\pi} \int_0^{\pi} V_m \sin \theta dt$$

$$= \frac{1}{\pi} \int_0^{\pi} V_m \sin \omega t dt$$

$$= \frac{V_m}{\pi} [-\cos \omega t]_0^{\pi}$$

$$= \frac{V_m}{\pi} (-\cos \theta)_0^{\pi} \quad [\theta = \omega t]$$

$$\begin{aligned}
 &= \frac{V_m}{\pi} [-\cos \pi + \cos \theta] \\
 &= \frac{V_m}{\pi} [(-(-1) + 1)] = \frac{2V_m}{\pi} \\
 \text{Average Value} &= \frac{2V_m}{\pi}
 \end{aligned}$$

1. RMS Value

$$\begin{aligned}
 V_{RMS} &= \sqrt{\frac{1}{\pi} \int_0^{\pi} V^2 dt} \\
 &= \sqrt{\frac{1}{\pi} \int_0^{\pi} (V_m \sin \theta)^2 dt} \\
 V_{RMS}^2 &= \frac{1}{\pi} \int_0^{\pi} V_m^2 \sin^2 \theta d\theta \\
 &= \frac{V_m^2}{\pi} \int_0^{\pi} \left(\frac{1 - \cos 2\theta}{2} \right) d\theta \\
 &= \frac{V_m^2}{\pi} \left[\theta - \frac{\sin 2\theta}{2} \right]_0^{\pi} \\
 &= \frac{V_m^2}{2\pi} \left[\pi - \frac{\sin 2\pi}{2} \right] \\
 V_{RMS}^2 &= \frac{V_m^2 \cdot \pi}{2\pi} \\
 V_{RMS}^2 &= \sqrt{\frac{V_m^2}{2}} \\
 V_{RMS} &= \frac{V_m}{\sqrt{2}}
 \end{aligned}$$

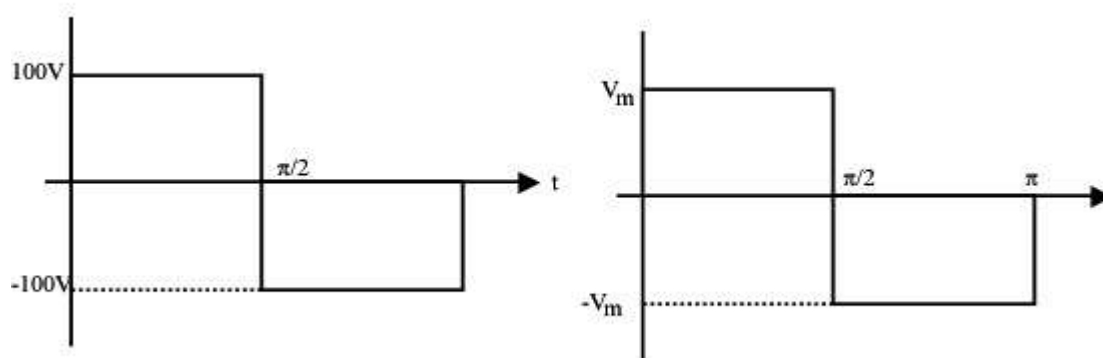
2. Form factor

$$\begin{aligned}
 &= \frac{\text{RMS Value}}{\text{Average Value}} \\
 &= \frac{\frac{V_m}{\sqrt{2}}}{\frac{2V_m}{\pi}} \\
 &= \frac{\pi}{2\sqrt{2}} = 1.110
 \end{aligned}$$

Peak factor

$$\begin{aligned}
 &= \frac{\text{Peak Value}}{\text{RMS Value}} \\
 &= \frac{V_m}{\frac{V_m}{\sqrt{2}}} \\
 &= \sqrt{2} = 1.414
 \end{aligned}$$

2.2 Calculate the form factor and peak factor



- It is a symmetrical wave

$$\begin{aligned}
 \text{Avg value} &= \frac{(\text{area under half cycle})}{\text{Base}} \\
 &= \frac{100 \times \frac{\pi}{2}}{\frac{\pi}{2}} = 100V
 \end{aligned}$$

$$\begin{aligned}
 \text{RMS value} &= \sqrt{\frac{\text{Squared area}}{\text{base}}} \\
 &= \sqrt{\frac{(l^2 * b)}{\text{base}}} \\
 &= \sqrt{\frac{100^2 * \frac{\pi}{2}}{\frac{\pi}{2}}}
 \end{aligned}$$

$$\text{RMS value} = 100V$$

$$\begin{aligned}
 &= \frac{l * b}{\frac{\pi}{2}} \\
 &= \frac{V_m * \frac{\pi}{2}}{\frac{\pi}{2}} \\
 \text{Avg value} &= V_m
 \end{aligned}$$

$$\begin{aligned}
 \text{RMS value} &= \sqrt{\frac{\text{Squared area}}{\text{base}}} \\
 &= \sqrt{\frac{(l^2 * b)}{\text{base}}} \\
 &= \sqrt{\frac{V_m^2 * \frac{\pi}{2}}{\frac{\pi}{2}}}
 \end{aligned}$$

$$\text{RMS value} = V_m$$

$$\begin{aligned}\text{Form factor} &= \frac{\text{RMS Value}}{\text{avg. value}} \\ &= \frac{100}{100}\end{aligned}$$

$$\text{Form Factor} = 1$$

$$\begin{aligned}\text{Peak factor} &= \frac{\text{Peak Value}}{\text{RMS value}} \\ &= \frac{100}{100}\end{aligned}$$

$$\text{Peak Factor} = 1$$

$$\begin{aligned}\text{Form factor} &= \frac{\text{RMS Value}}{\text{avg. value}} \\ &= \frac{V_m}{V_m}\end{aligned}$$

$$\text{Form Factor} = 1$$

$$\begin{aligned}\text{Peak factor} &= \frac{\text{Peak Value}}{\text{RMS value}} \\ &= \frac{V_m}{V_m} = 1\end{aligned}$$

$$\text{Peak Factor} = \frac{\text{Max. Value}}{\text{RMS value}} = \frac{V_m}{V_m} = 1$$

2.3 Derive the expression for R.M.S. value, Average value, Form factor, Peak factor of half wave rectified waveform is shown in below figure.

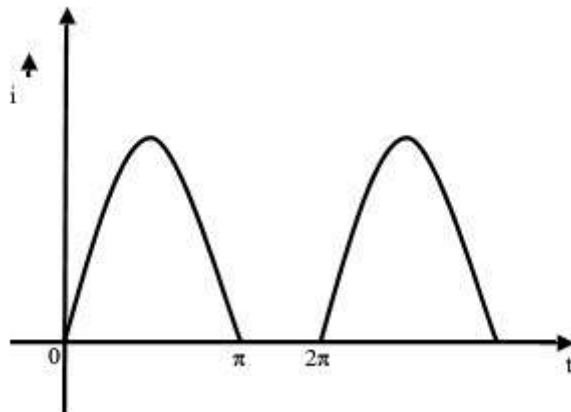
$$i = I_m \sin \theta$$

$$i = 0$$

$$0 < \theta \leq \pi$$

$$\pi \leq \theta \leq 2\pi$$

To find the RMS value



$$\begin{aligned}
 I_{RMS} &= \sqrt{\int_0^{2\pi} \frac{i^2 d\theta}{2\pi}} \\
 &= \sqrt{\int_0^{\pi} \frac{i^2 d\theta}{2\pi} + \int_{\pi}^{2\pi} \frac{i^2 d\theta}{2\pi}} \\
 &= \sqrt{\int_0^{\pi} \frac{I_m^2 \sin^2 \theta d\theta}{2\pi} + 0} \\
 &= \sqrt{\frac{I_m^2}{2\pi} \int_0^{\pi} \sin^2 \theta d\theta} \\
 &= \sqrt{\frac{I_m^2}{2\pi} \int_0^{\pi} \frac{1 - \cos 2\theta d\theta}{2}} \\
 &= \sqrt{\frac{I_m^2}{4\pi} \int_0^{\pi} d\theta - \frac{I_m^2}{4\pi} \int_0^{\pi} \cos 2\theta d\theta} \\
 &= \sqrt{\frac{I_m^2}{4\pi} [\theta]_0^{\pi} - \frac{I_m^2}{4\pi} \left[\frac{\sin 2\theta}{2} \right]_0^{\pi}} \\
 &= \sqrt{\frac{I_m^2}{4\pi} [\pi - 0] - 0} = \sqrt{\frac{I_m^2}{4}}
 \end{aligned}$$

$$\therefore I_{RMS} = \frac{I_m}{2}$$

To find the average value

$$I_{av} = \int_0^{2\pi} \frac{id\theta}{2\pi} \quad (\text{since it is an unsymmetrical wave we have to consider})$$

$$\begin{aligned}
 I_{av} &= \int_0^{\pi} \frac{id\theta}{2\pi} + \int_{\pi}^{2\pi} \frac{id\theta}{2\pi} \\
 &= \int_0^{\pi} \frac{I_m}{2\pi} \sin \theta d\theta + 0 \\
 &= \frac{I_m}{2\pi} \int_0^{\pi} \sin \theta d\theta \\
 &= \frac{I_m}{2\pi} [-\cos \theta]_0^{\pi} = \frac{I_m}{2\pi} [-\cos \pi + \cos \theta] = \frac{I_m}{2\pi} \cdot 2
 \end{aligned}$$

$$I_{av} = \frac{I_m}{\pi}$$

$$\text{Form factor} = \frac{\text{RMS Value}}{\text{Average Value}} = \frac{I_m/2}{I_m/\pi} = \frac{\pi}{2} = 1.57$$

$$\text{Peak factor} = \frac{\text{Maximum Value}}{\text{RMS Value}} = \frac{I_{\max}}{I_{\max}/2} = 2$$

2.4 The equation of an alternating current is given by $i = 40\sin 314t$. Determine the following

- i. Maximum value of current
- iii. R.M.S. value of current
- v. Form factor

- ii. Average value of current
- iv. Frequency and angular frequency
- vi. Peak factor

Solution

Given

$$i = 40\sin 314t \quad (i)$$

We have

$$i = I_m \sin \omega t \quad (ii)$$

Comparing (i) and (ii) we get

$$\text{Maximum value of current } I_m = 40A$$

$$\text{Angular frequency } \omega = 314 \text{ rad / second}$$

$$\begin{aligned} \text{Average value } I_{av} &= \frac{2I_m}{\pi} = \frac{2 \times 40}{\pi} \\ &= 25.48A \end{aligned}$$

$$\text{RMS value } I_{RMS} = \frac{I_{\max}}{\sqrt{2}} = 28.28A$$

$$\begin{aligned} \text{Frequency } f &= \frac{\omega}{2\pi} \\ &= \frac{314}{2\pi} = 50 \text{ Hz} \end{aligned}$$

$$\text{Form factor} = \frac{\text{RMS Value}}{\text{Average Value}} = \frac{28.288}{25.464} = 1.11$$

$$\text{Peak factor} = \frac{\text{Maximum Value}}{\text{RMS Value}} = \frac{40}{28.288} = 1.414$$

2.5 What is the equation of 50 Hz frequency sine wave with an RMS value of 50V.

Solution:

Given,

$$f = 50 \text{ Hz}$$

$$V_{R.M.S.} = 50V$$

We have,

$$v = V_{\max} \sin \omega t$$

Where $V_{MAX} = V_{R.M.S} \times \sqrt{2}$ and
 $\omega = 2\pi f$
 $\omega = 314 \text{ rad/s}$

$\therefore$ Voltage equation is given by

$$V = 70.7 \sin(314)t$$

2.6 An alternating voltage of frequency 50 Hz has an R.M.S. value of 200 V. Find the instantaneous value of voltage after 2.5 milliseconds. Also determine the time for the instantaneous voltage of 150 V.

Solution:

Given,

$$\text{Frequency (f)} = 50 \text{ Hz}$$

$$V_{R.M.S} = 200 \text{ V}$$

$$\begin{aligned} V_{MAX} &= 200 \times \sqrt{2} \\ &= 282.84 \text{ V} \end{aligned}$$

The equation for voltage is given by,

$$\begin{aligned} v &= V_{MAX} \sin \omega t \\ &= 282.2 \sin(2\pi \times 50t) \\ &= 282.8 \sin 314t \end{aligned}$$

To find the instantaneous value of voltage after 2.5 milliseconds

$$\begin{aligned} v &= 282.8 \sin 314 \times (2.5 \times 10^{-3}) \\ &= 282.8 \sin(0.785) \quad [\text{In Radian Mode}] \\ &= 199.89 \text{ volts} \end{aligned}$$

$$\begin{aligned} \text{(or)} \quad v &= 282.8 \sin(2\pi \times 50)t \\ &= 282.8 \sin(2 \times 180 \times 50) \times 2.5 \times 10^{-3} \\ &= 282.8 \sin(45^\circ) \end{aligned}$$

To find the time to reach the instantaneous voltage of 150 V

$$v = V_{\text{MAX}} \sin(2\pi f)t$$

$$150 = 282.8 \sin(2 \times 180 \times 50)t$$

$$\frac{150}{282.8} = \sin(18000t)$$

We have,

$$18000t = \sin^{-1}\left(\frac{150}{282.8}\right)$$

$$t = \frac{32.033}{18000}$$

$$t = 1.779 \times 10^{-3} \text{ seconds}$$

2.7 Determine the average value and the R.M.S. value of the triangle wave as show in the below figure.

Solution

The coordinates of 'O' are

$$O(0,0) \rightarrow x_1 = 0; y_1 = 0$$

$$V_{\text{RMS}} = \sqrt{\frac{\text{Squared area}}{\text{base}}}$$

$$= \sqrt{\frac{\frac{1}{2}bh^2}{\text{base}}}$$

$$= \sqrt{\frac{\frac{1}{2} \pi \cdot v_{\text{max}}^2}{\pi}}$$

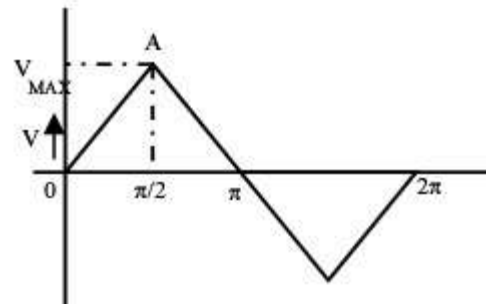
$$= \frac{V_m}{\sqrt{2}}$$

$$= \sqrt{\frac{8}{\pi^3} \cdot V_{\text{max}}^2 \left[\frac{\theta^3}{3} \right]_0^{\pi/2}}$$

$$= \sqrt{\frac{8V_{\text{max}}^2}{\pi^3 \times 3} \left[\frac{\pi^3}{8} - 0 \right]}$$

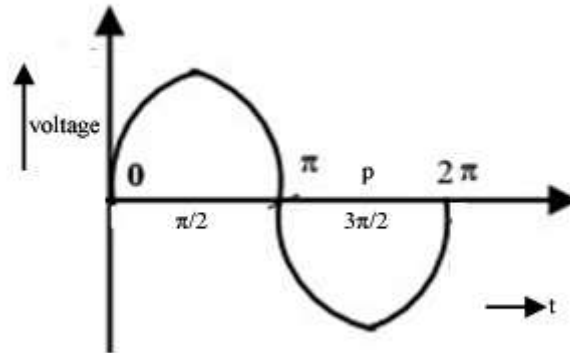
$$= \sqrt{\frac{V_{\text{max}}^2}{3}}$$

$$\therefore V_{\text{R.M.S}} = \sqrt{\frac{V_{\text{max}}^2}{3}}$$

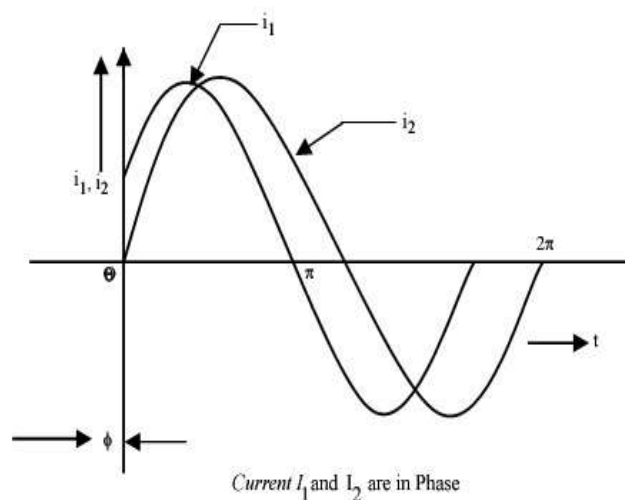


2.1.12 PHASE

The phase of an alternating quantity at point 'p' shown in the figure below implies that the phase is nothing but a fraction of time period that has elapsed position (is this zero position). Two alternating quantities are said to be reach their zero value or maximum value at the same time.



Phase representation of sinusoidal wave



For example the two alternating quantities namely.

$$i_1 \rightarrow I_{m1} \sin \theta$$

$$i_2 \rightarrow I_{m2} \sin \theta$$

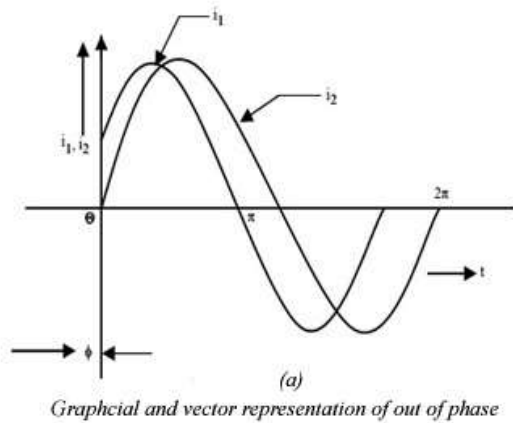
From the above figure it is clear that both i_1 and i_2 reach their zero (or) maximum same time and hence are said to be in phase. Note that phase difference.

Out of phase or phase difference

When the two alternating quantities do not reach their zero or maximum value time, then they are said to be out of phase. This difference is measured in degrees or in radians. It is given in terms of 'lead' or 'lag'.

Leading alternating quantity is one which reaches its maximum or zero than the other alternating quantity.

$$\begin{aligned} \text{if } i_1 &= I_{m1} \sin(\omega t + \phi) \\ i_2 &= I_{m2} \sin(\omega t) \end{aligned}$$



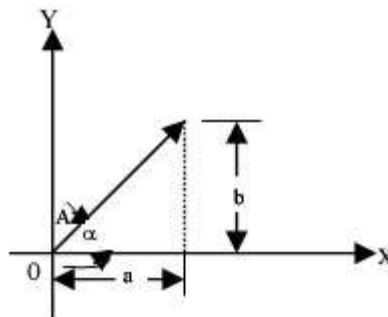
2.1.13 COMPLEX ALGEBRA

A vector quantity has both magnitude and direction whereas a scalar quantity has only magnitude but no direction. A vector can be written in terms of its horizontal components. In the below figure 'OM' represents the vector 'A' and is represented by the 'a' and 'b' respectively. Thus $A = a + jb$ where 'j' is an operator indicating that 'b' is perpendicular to 'a'. This form of representing a vector is known as *complex form*.

'a' is the in phase (or) active component of vector A.

'b' is the quadrature (or) reactive component of vector A.

The magnitude of vector A is given by $(A) = \sqrt{a^2 + b^2}$
 the angle vector 'A' with x-axis is given by $\tan^{-1} b/a$.



Vector representation for vertical and horizontal complex vector

Note:

1. $j = \sqrt{-1}$. It also indicates counter clock-wise rotation of vector
2. $j^2 = j \cdot j = -1$. It also indicates counter clock-wise rotation of vector
3. $j^3 = j \cdot j \cdot j = -j$. It also indicates counter clock-wise rotation of vector
4. $j^4 = j \cdot j \cdot j \cdot j = j^2 \cdot j^2 = (-1)(-1) = 1$. It also indicates counter clock-wise rotation of vector through 360°
5. $\frac{1}{j} = \frac{1 \cdot j}{j \cdot j} = \frac{j}{-1} = -j$.
6. $-j$ indicates clock wise rotation of vector through 90°

Vector 'A' written in the form of $5 + j7$ is known as rectangular form. The magnitude of $A = |A| = \sqrt{5^2 + 7^2} = 8.6$

Polar representation, from the above figure,

$$A = |A| \angle \alpha^\circ$$

$$|A| = \sqrt{a^2 + b^2}$$

$$\tan \alpha = \frac{\sin \alpha}{\cos \alpha} = \frac{b/m}{a/m} = \frac{b}{a}$$

$$\alpha^\circ = \tan^{-1} \frac{b}{a}$$

Rectangular representation, from the above figure,

$$\sin \alpha = \frac{\text{opposite side}}{\text{Hypotenuse}} = \frac{b}{M}$$

$$b = M \sin \alpha$$

$$\cos \alpha = \frac{\text{Adjacent side}}{\text{Hypotenuse}} = \frac{a}{M}$$

$$a = M \cos \alpha$$

The angle $\alpha = \tan^{-1}(7/5) = 54.46^\circ$

The same vector can be written as

$$A = |A| \angle \alpha^\circ$$

$$\text{i.e., } A = 8.6 \angle 54.46^\circ.$$

This form of representation of vector is known as polar form.

Note:

1. Addition and subtraction of vector can be easily done in rectangular representation.
2. Multiplication and division of vector can be easily done in polar representation.

Problems

2.8 Let $Z = (10 + j15)\Omega$. Convert this into polar form.

Note:- The author had used the scientific CASIO fx 991MS.

Solution:

Steps

1. Press Shift key
2. Press +
3. Display POL(
4. Press 10, 15) and =. It shows $18.03\ \Omega$
5. Press ALPHA F. It shows the angle 56.31°
6. In the polar form, now Z is written as $Z = 18.03\angle 56.31^\circ$

2.9 Convert $20 - j15$ into polar form.

Solution:

Steps

1. Press Shift key
2. Press +
3. Display POL(
4. Press 20, - 15) and =. It shows 25Ω
5. Press ALPHA F. It shows the angle -36.9°
6. In the polar form, now Z is written as $Z = 25\angle -36.9^\circ$

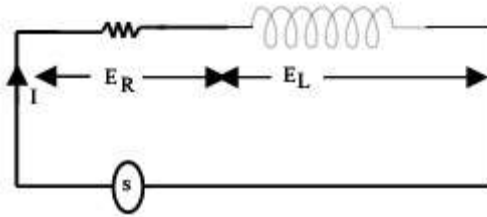
2.10 Convert $Z = 30\angle -45^\circ$ into rectangular form.

Solution:

Steps

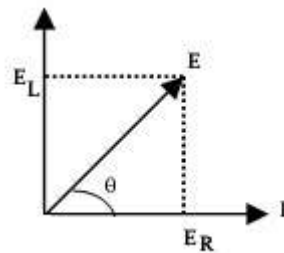
1. Press Shift key
2. Press -
3. Display Rec(
4. Press 30, - 45) and =. It shows 21.21Ω (active part)
5. Press ALPHA F. It shows the -36.9°
6. In the Polar form, now Z is written as $Z = 21.21 - j21.22$

2.2 Series RLC Circuits**2.2.1RL circuit**



Consider the RC circuit, Let the applied voltage be 'E' volt and the current flowing through it 'I' amps and E_R , E_L are the voltage drop across the individual elements.

Phase diagram



$$\text{Apparent power} = EI$$

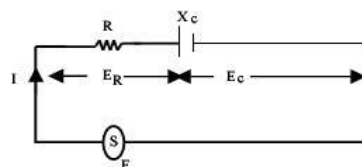
$$\text{True / Real power} = EI \cos \theta$$

$$\cos \theta = \frac{R}{|Z|}$$

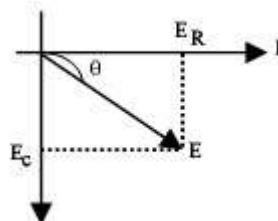
$$\text{Impedance } Z = R + jX_L$$

$$X_L = \omega L$$

2.2.2 RC circuit



Consider the R_L series circuit. The phasor diagram is given by



$$\text{Apparent power} = EI$$

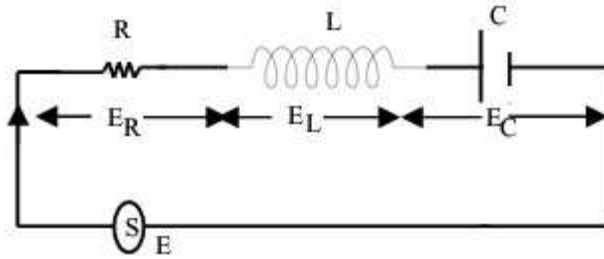
$$\text{Real power} = EI \cos \theta$$

$$\cos \theta = \frac{R}{|Z|}$$

$$\text{Impedance } Z = R - jX_C$$

$$X_C = \frac{1}{\omega C}$$

2.2.3 RLC series circuit



Consider RLC series circuit let the voltage applied be 'E' and an voltage drop across R be E_R , L be E_L and C be E_C .

The voltage across resistance $E_R = IR$

The voltage across inductance $E_L = I \times L$

The voltage across resistance $E_C = I \times C$

$$\begin{aligned} \text{The net voltage } E &= E_R + jE_L - jE_C \\ &= IR + jIX_L - jIX_C \\ &= I[R + jX_L - jX_C] \\ &= I[R + j(X_L - X_C)] \\ &\quad \underbrace{\hspace{1.5cm}}_Z \end{aligned}$$

$$E = IZ$$

Where voltage $E = IZ$

current $I = E/Z$

voltage $Z = R + j(X_L - X_C)$

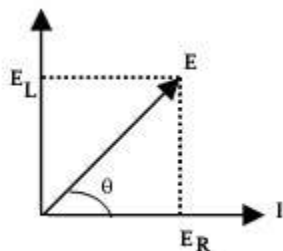
$$X_L = \omega L; \quad X_C = \frac{1}{\omega C}$$

$$|Z| = \sqrt{R^2 + (X_L - X_C)^2};$$

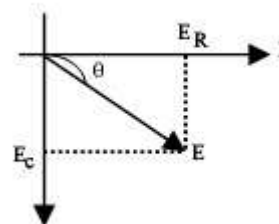
$$\theta = \tan^{-1} \left(\frac{X_L - X_C}{R} \right)$$

Phasor diagrams

1. If $X_L > X_C$ then the ckt behaves like R-L ckt

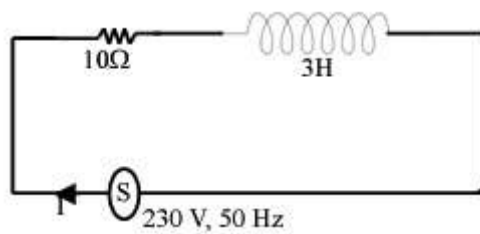


- If $X_C > X_L$ then the ckt behaves like R-C ckt



Problems

2.11 Calculate power factor, impedance, circuit and power of the given R_L circuit. The resistance is $10\ \Omega$, Inductance is $3H$ and voltage supply across the circuit



Given:

$$R = 10\Omega$$

$$L = 3H$$

$$V = 230V$$

$$f = 50Hz$$

To find: Impedance, current, power factor, power.

Solution:

$$\begin{aligned} X_L &= \omega L \\ &= 2\pi fL \\ &= 2 \times 3.14 \times 50 \times 3 \\ X_L &= 942 \end{aligned}$$

1. Impedance

$$\begin{aligned} Z &= R + jX_L \\ &= 10 + j942 \\ Z &= 942.05 \angle 89.93 \approx 942.05 \angle 90 \end{aligned}$$

2. Current

$$\begin{aligned} I &= \frac{V}{Z} \\ &= \frac{230}{942} = 0.244 \text{ A} \end{aligned}$$

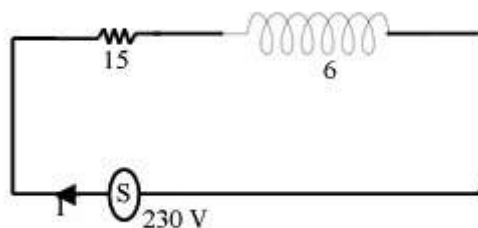
3. Power factor

$$\begin{aligned} \cos \phi &= \frac{R}{Z} \\ &= \frac{10}{942} = 0.01 \end{aligned}$$

4. Power

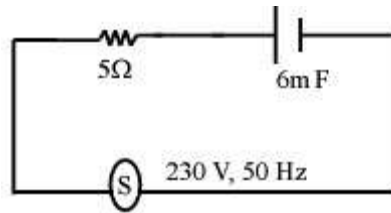
$$\begin{aligned} \text{power} &= VI \cos \phi \\ &= 230 \times 0.244 \times 0.01 \\ &= 0.57 \text{ W} \end{aligned}$$

2.12 Calculate the impedance, power factor, current power in the following circuit.



(Answers: $Z = 1884 \angle 90^\circ$; $I = 0.122 \text{ A}$; $\cos \phi = 7.96 \times 10^{-3}$ power = 0.22 watts)

2.13 Calculate impedance z, power p and power factor for the following circuit

**Given:**

$$R = 5\Omega$$

$$C = 6\mu F = 6 \times 10^{-6} F$$

$$V = 230V$$

$$f = 50Hz$$

Solution:

$$\begin{aligned} X_C &= \frac{1}{\omega_L} \\ &= \frac{1}{2\pi fC} \\ &= \frac{1}{2 \times 3.14 \times 50 \times 6 \times 10^{-6}} \\ X_C &= 530.78 \end{aligned}$$

i. Impedance

$$\begin{aligned} Z &= R - jX_C \\ &= 5 - j(530.78) \\ Z &= 230.80 \angle 90^\circ \end{aligned}$$

2. Current

$$\begin{aligned} I &= \frac{V}{Z} \\ &= \frac{230}{530} = 0.43A \end{aligned}$$

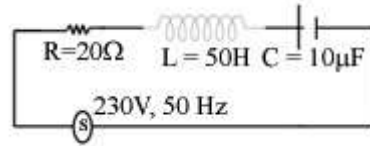
3. Power factor

$$\begin{aligned} \cos \phi &= \frac{R}{Z} \\ &= \frac{5}{531} = 0.009 \end{aligned}$$

4. Power

$$\begin{aligned}
 \text{power} &= VI \cos \phi \\
 &= 230 \times 0.43 \times 0.009 \\
 &= 0.9 \text{ Watts}
 \end{aligned}$$

2.14 Calculate the impedance, current for the given RLC circuit $R = 20\Omega$, $L = 50 \text{ H}$, $C = 10\mu\text{F}$, $V = 230\text{V}$, $f = 50\text{Hz}$



Given:

$$R = 20\Omega$$

$$L = 50\text{H}$$

$$C = 10\mu\text{F} = 10 \times 10^{-6} \text{ F}$$

$$V = 230\text{V}$$

$$f = 50\text{Hz}$$

Solution:

$$\begin{aligned}
 X_L &= \omega \times L \\
 &= 2\pi f \times L \\
 &= 2 \times \pi \times 50 \times 50 \\
 &= 15700
 \end{aligned}$$

$$\begin{aligned}
 X_C &= \frac{1}{\omega C} \\
 &= \frac{1}{2\pi f \times C} \\
 &= \frac{1}{2 \times \pi \times 50 \times 10 \times 10^{-6}}
 \end{aligned}$$

$$X_C = 318.47$$

$$(X_C - X_L) = 15700 - 318.47$$

$$(X_C - X_L) = 15381.52$$

i. Impedance

$$\begin{aligned}
 Z &= R + j(X_L - X_C) \\
 &= 20 + j(15381.52)
 \end{aligned}$$

$$Z = 1581.53 \angle 89.9^\circ$$

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$Z = 15301.54$$

ii. Current

$$I = \frac{V}{Z}$$

$$I = \frac{230}{15381.54} = 0.014A$$

$$\phi = \tan^{-1} \frac{(X_L - X_C)}{R}$$

$$= \tan^{-1} \frac{(15381.53)}{20}$$

$$\phi = 89.9$$

iii. Power factor

$$\cos \phi = 1.30 \times 10^{-3}$$

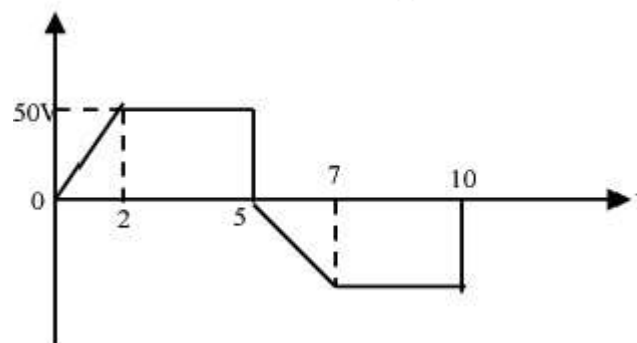
iv. Power

$$power = VI \cos \phi$$

$$= 230 \times 0.014 \times 1.30 \times 10^{-3}$$

$$= 4.186 \times 10^{-3} W$$

2.15 Find the average value of the voltage waveform shown in the figure



Vavg. of triangle

$$= \frac{\frac{1}{2}bh}{base}$$

$$= \frac{\frac{1}{2} \times 2 \times 50}{2}$$

$$= 25$$

Vavg. of rectangle

$$= \frac{l \times b}{base}$$

$$= 3$$

$$= 50$$

$$\begin{aligned}
 V_{avg} &= V_{avg\Delta} + V_{avg\square} \\
 &= 25 + 50 \\
 &= 75
 \end{aligned}$$

$$\begin{aligned}
 \text{Average Value} &= \frac{V_{avg}}{\text{no. of shapes}} \\
 &= \frac{75}{2} \\
 &= 37.5V
 \end{aligned}$$

2.3 Power

2.3.1 Reactive Power

Reactive power is needed in an alternating current transformation system to support the transfer of real power over the network (i.e.) the portion of power flow that is temporarily stored in the form of magnetic or electric field, due to inductive and capacitive network elements and then return to source is known as reactive power.

It is represented as Q

Its Unit is V_{Ar} (Volt ampere reactive)

2.3.2 Real Power

The portion of power flow remaining after being averaged over a complete AC waveform is real power (i.e.) the energy that can be used to work is called real power. It is also called as active power.

It is represented as P

Its Unit is KW (Kilo Watts)

2.3.3 Complex Power

The Vector sum of real and reactive power is known as complex power.

It is given by (P + jQ)

P – real power

Q – reactive power

It is denoted as 's' and its unit is VA

2.3.4 Apparent Power

It is the magnitude of complex power.

(i.e.,) $s = |P + jQ|$ unit is VA.

2.4 Introduction to 3 ϕ system

The generation, transmission, distribution and utilization of large blocks of electric power are accomplished by means of three phase circuit.

A balanced three phase system consists of three sinusoidal voltages of identical amplitudes and frequency but are out of phase with each other by 120° .

A standard practice is to refer the three phase system A, B, C (or) 'A' phase is most used as reference phase. B and C phases are out of phase with 'A' by 120° and 240° respectively.

$$V_a = V \angle 0^\circ$$

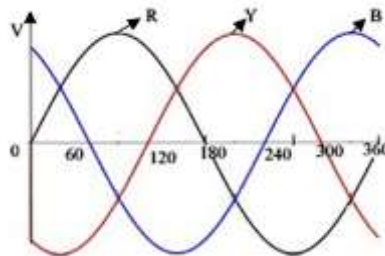
$$V_b = V \angle 120^\circ$$

$$V_c = V \angle 240^\circ$$

$$V_a, V_b, V_c \rightarrow \text{Phase Voltage}$$

$$V \rightarrow \text{Supply Voltage}$$

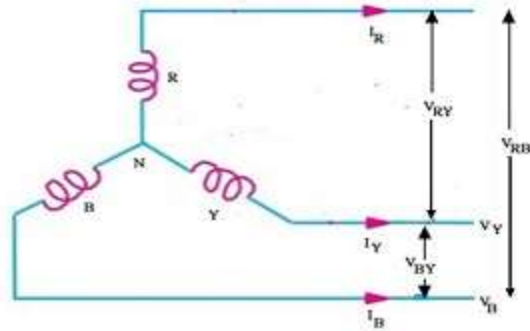
An important characteristics of balanced three phase voltage is that sum of the voltage is zero ($V_a + V_b + V_c = 0$)



The three phase system can be either Y (star) or Δ (Delta) connected.

Star Connection

- it is a 3 ϕ , 3 wire system (i.e.) R, Y, B phase and a neutral to ground (or)
- It is a 3 ϕ , 4 wire system (i.e.) R, Y, B phase and a neutral to load.
- In star connected supply, the line current = phase current and line voltage $\neq$ phase voltage.



Phase Voltage

The voltage between any one of the conductors in a three phase system to ground.

$$(V_{AN} + V_{BN} + V_{CN} = 0) \quad (N - \text{ground/neutral})$$

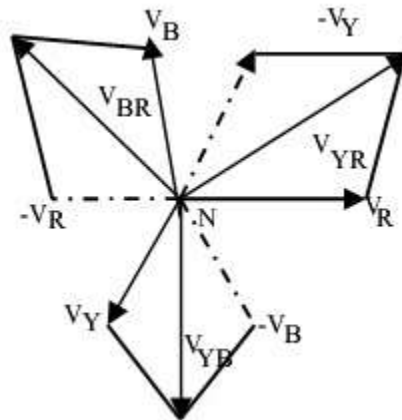
Line Voltage

The voltage presents between any two of the conductors in a three phase system. It is 1.732 times higher than phase voltage in a star connection.

$$V_{AB} + V_{BC} + V_{CA}$$

Phasor Diagram

For star connected system, the voltage phasor diagram is given by



Consider the phasor diagram

$$V_{RY} = V_R + (-V_Y)$$

$$V_{RY} = V_R - V_Y$$

From vector diagram

- The phase angle between V_R and $-V_Y = 360^\circ$
- By parallelogram law of vectors.

$$V_{RY} = \sqrt{V_R^2 + (-V_Y)^2 + 2V_R V_Y \cos 60^\circ}$$

In star connection $V_R = V_Y = V_B = V_{ph}$ (phase voltage)

$$\begin{aligned} V_{RY} &= \sqrt{V_{ph}^2 + V_{ph}^2 + 2V_{ph}^2 \cos 60^\circ} \\ &= \sqrt{V_{ph}^2 + V_{ph}^2 + 2V_{ph}^2 \left(\frac{1}{2}\right)} \\ &= \sqrt{3V_{ph}^2} = \sqrt{3}V_{ph} \end{aligned}$$

Since $V_{RY} = V_{YB} = V_{BR} = V_{l-l}$ (line voltage)

$$V_{l-l} = \sqrt{3}V_{ph}$$

and $V_{l-l} = I_{ph}$

3 ϕ Power (Expression for power) (Jan: 2013 Part-A)

3 ϕ apparent power = 3 $\times$ apparent power per phase

$$= 3 \times V_p I_p$$

In terms of line parameters

$$\begin{aligned} &= 3 \times \frac{V_L}{\sqrt{3}} \times I_L \quad (I_L = I_p) \\ &= \sqrt{3} \times V_L \times I_L \text{ (VA)} \end{aligned}$$

3 ϕ Real Power

3 ϕ Real power = 3 $\times$ Real power per phase

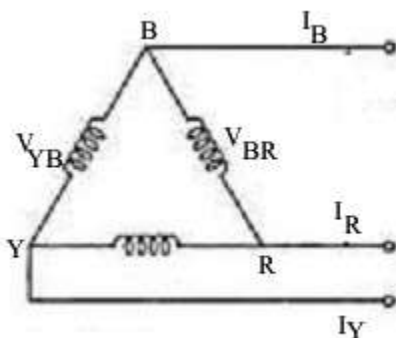
Real power o/p per phase = $V_p I_p \cos \phi$. ($\cos \phi$ = power factor)

3 ϕ Real power = 3 $\times V_p \times I_p \times \cos \phi$

$$= 3 \times \frac{V_L}{\sqrt{3}} \times I_L \times \cos \phi$$

$$= \sqrt{3} V_L I_L \cos \phi. \text{ (W)}$$

Mesh or Delta System



- There is no neutral point.

- Line current and phase current are different.

$$(i.e.) I_L \neq I_{ph}$$

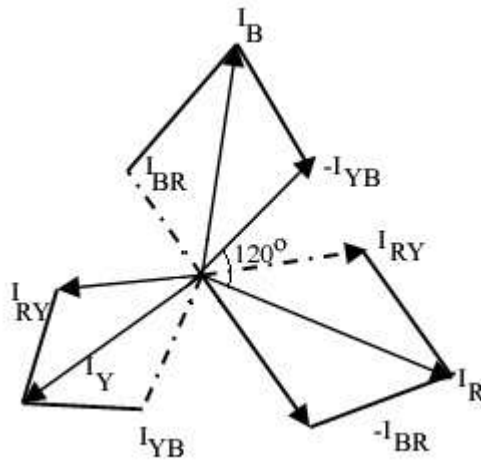
- Line voltage and line current are same

$$(i.e.) I_L = I_{ph}$$

$I_R, I_Y, I_B \rightarrow$ line current

$I_{RY}, I_{BR}, I_{YB} \rightarrow$ phase current

Phasor diagram



From phasor diagram

$$\text{line current } I_R = I_{RY} - I_{BR}$$

From the vector diagram: I_{RY} and I_{BR} has phase diff = 360°

By parallelogram law of vectors

$$I_R = \sqrt{I_{RY}^2 + I_{BR}^2 + 2I_{RY}I_{BR} \cos 60^\circ}$$

Assuming balanced load

$$(i.e.) I_{RY} = I_{BR} = I_{ph}$$

$$I_R = \sqrt{I_p^2 + I_p^2 + 2I_pI_p \cos 60^\circ}$$

$$= \sqrt{I_p^2 + I_p^2 + 2I_p^2 \left(\frac{1}{2}\right)}$$

$$= \sqrt{3I_p^2}$$

$$I_R = \sqrt{3}I_p$$

where I_R - line current

$$I_L = \sqrt{3}I_p$$

and $V_L = V_p$

$\therefore$ In delta system line voltage = phase voltage and line current $\sqrt{3}$ phase current.

3 ϕ Power (real power)

3 ϕ Real power = 3 $\times$ Real power per phase

$$= 3 \times V_p \times I_p \cos \phi.$$

$$= 3 \times V_L \times \frac{I_L}{\sqrt{3}} \times \cos \phi$$

$$= \sqrt{3} V_L I_L \cos \phi. \quad (\text{W})$$

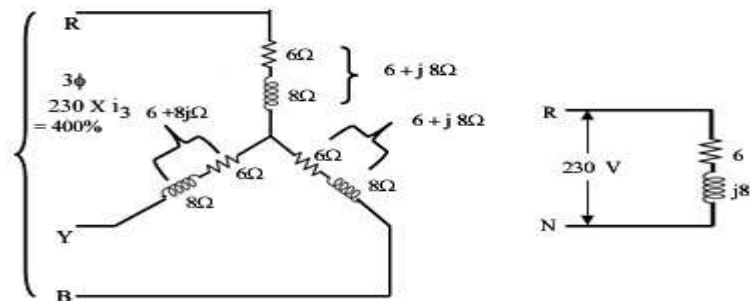
Advantages of 3 ϕ system over 1 ϕ system: (Part – A April 2010)

- Universally adapted for generation, transmission, distribution of electric power.
- 3 ϕ is more capable and reliable than 1 ϕ system.
- Parallel operation of 3 ϕ system is simple when compared to 1 ϕ system (alternator).
- 3 ϕ system produces higher output than 1 ϕ system.
- 3 ϕ motors have uniform torque where as 1 ϕ motor have pulsating torque.

Problems

2.16 A balanced three phase star connected load consists of 6 Ω resistor and 8 Ω reactor (inductive) in each phase the supply is 230V 3 ϕ 50 Hz. Find (Part –II April 2010)

- a) The phase current
- b) line current
- c) total power



- a) Phase current

$$I_{ph} = \frac{V}{Z}$$

$$= \frac{230}{6 + j8} = \frac{230 \angle 0^\circ}{10 \angle 53.13^\circ} = 23 \angle -53.13^\circ \text{ A}$$

$$I_{line} = I_{phase} = 23 \angle -53.13^\circ \text{ A}$$

$$mag = 23 \text{ A}$$

- b) Line current = 23 A

- c) Power: ?

Power factor = ?

Phase angle = Angle between supply voltage and current

$$\phi = -53.13^\circ$$

Power Factor:

$$= \cos \phi$$

$$= \cos(-53.13^\circ)$$

$$= \cos(53.13^\circ)$$

$$= 0.600$$

Total power:

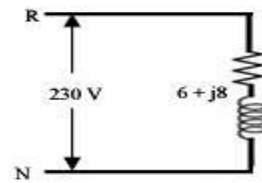
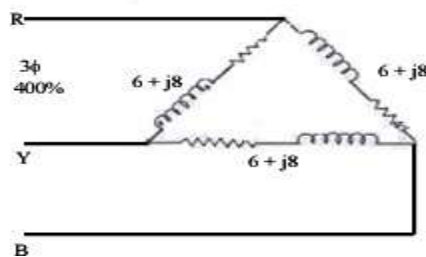
$$= 3 \times V_{ph} \times I_{ph} \cos \phi$$

$$= 3 \times 230 \times 23 \times 0.6$$

$$= 9522W$$

$$= 9.5KW$$

2.17 A balanced 3 ϕ delta connected load consists of 6Ω resistor and 8Ω reactor (inductive) in each phase. The supply is 230V, 3 ϕ , 50Hz, Find a) phase current, b) line current, c) power.



a) Phase current

$$I_{ph} = \frac{V}{Z}$$

$$= \frac{400}{6 + j8} = \frac{400 \angle 0^\circ}{10 \angle 53.13^\circ} = 40 \angle -53.13^\circ A$$

b) I_{line}

$$I_{line} = \sqrt{3} I_{ph}$$

$$= \sqrt{3} \times 40$$

$$I_{line} = 69.3A$$

$$V_{line} = V_{phase}$$

Power Factor:

$$= \cos \phi$$

$$= \cos(-53.13^\circ)$$

$$= \cos(53.13^\circ)$$

$$= 0.6 \quad (\text{lag})$$

Total power:

$$\begin{aligned}
 &= 3 \times V_{ph} \times I_{ph} \cos \phi \\
 &= 3 \times 400 \times 40 \times 0.6 \\
 &= 28800W \\
 &= 28.8KW
 \end{aligned}$$

2.18 Each phase of a 3Ω alternator produces a voltage of 6351 V and carry a max., current of 315A. Find line voltage, max line current and KVA if it is connected in a star and delta connection

a) star connection

$$\begin{aligned}
 V_L &= \sqrt{3} \times V_{ph} \\
 &= \sqrt{3} \times 6.351 \\
 &= 11000V \\
 I_L &= I_{ph} = 351A \\
 \text{Total capacity} &= \sqrt{3} \times V_L I_L \\
 &= \frac{\sqrt{3} \times 11,000 \times 315}{1000} \text{ KVA} \\
 &= 6000\text{KVA}
 \end{aligned}$$

b) delta connection

$$\begin{aligned}
 V_L &= V_{ph} = 6,351V \\
 I_L &= \sqrt{3} I_{ph} = \sqrt{3} \times 351A \\
 &= 545.6A \\
 \text{Total capacity} &= \frac{\sqrt{3} \times 6351 \times 545.6}{1000} \text{ KVA} \\
 &= 6,000\text{KVA}
 \end{aligned}$$

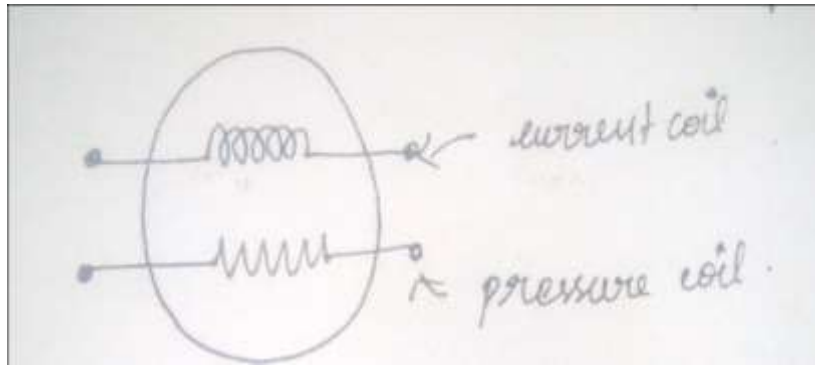
2.5 Power measured by two wattmeter method (Part B Jan 2013)

Wattmeter

Wattmeter is an electrical device which is used to sense voltage, current and phase angle between voltage and current and shows the power in watts.

It consists of 2 coils:-

- current coil
- pressure coil



Current coil

Current coil senses the current and is always connected in series with load. Current coil resistance is low and hence it has less number of turns and large cross sectional area.

Pressure coil

Pressure coil (or) voltage coil senses the voltage and is always connected across the supply. The resistance of this coil is very high and hence it has larger number of turns and small cross section area.

The total power developed in the 3ϕ balanced load is given by

$$P = 3VI \cos \phi$$

Where $VI \cos \phi$ – wattmeter reading

$\cos \phi$ – power factor

W – wattmeter reading

V – voltage across pressure coil

I – current through current coil

ϕ – phase angle difference between voltage and current

Power measurement by two wattmeter method

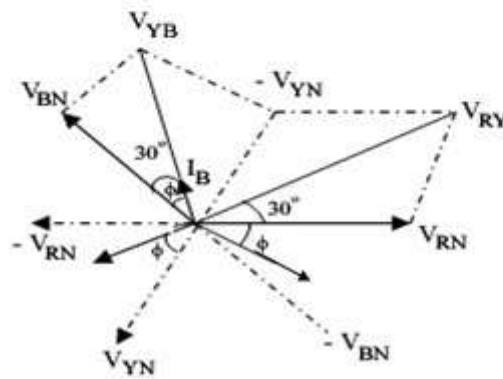
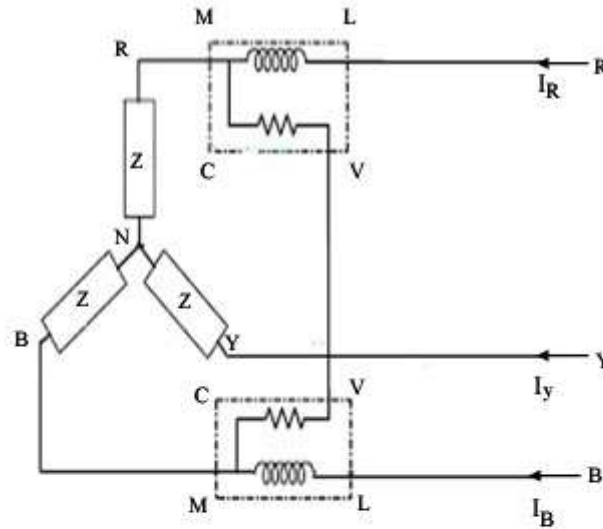
Two wattmeter method overcome this disadvantage of three wattmeter power measurement.

Number of wattmeter required to measure the total power is reduced. This method is applied usually for measuring the electric power in 3ϕ , 3 wire circuit.

The load may be balanced (or) unbalanced. It may be either connected in delta (or) star.

Star connected load

In a star connected system the two wattmeter current coils are inserted in two of three lines and voltage coils to the line in which wattmeter is connected.



Phaseor Diagram

Let V_{RN} , V_{YN} , V_{BN} be the RMS value of the 3ϕ load phase voltage and I_R , I_Y , I_B be the RMS value of phase current

Consider the wattmeter W_1

- current through current coil of $W_1 = I_R$.
- Potential difference across potential coil of $W_1 = V_{RY}$

$$V_{RY} = V_{RN} - V_{YN}$$
- Phase angle between V_{RY} and I_R is $(30^\circ + \phi)$

$$\therefore W_1 = V_{RY} I_R \cos(30^\circ + \phi)$$

Since the load is balanced $V_{RY} = V_L$ and $I_R = I_L$

$$W_1 = V_L I_L \cos(30^\circ + \phi) \quad (1)$$

Consider the wattmeter W_2 ,

- current through current coil $W_2 = I_B$.
- Potential difference between across potential coil of $W_2 = V_{BY}$.

$$V_{BY} = V_{BN} - V_{YN}$$
- The phase angle between V_{BY} and I_B is $(30^\circ - \phi)$

$$W_2 = V_L I_L \cos(30^\circ + \phi) \quad (2)$$

$$\text{Total Power } P = W_1 + W_2$$

$$\begin{aligned} P &= V_L I_L \cos(30^\circ + \phi) + V_L I_L \cos(30^\circ - \phi) \\ &= V_L I_L [\cos(30^\circ + \phi) + \cos(30^\circ - \phi)] \quad (3) \end{aligned}$$

Formula:

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\cos(A - B) = \cos A \cos B + \sin A \sin B$$

Applying the formula in equation (3)

$$\begin{aligned} P &= V_L I_L [(\cos 30 \cos \phi - \sin 30 \sin \phi) + (\cos 30 \cos \phi + \sin 30 \sin \phi)] \\ &= V_L I_L [\cos 30 \cos \phi - \sin 30 \sin \phi + \cos 30 \cos \phi + \sin 30 \sin \phi] \\ &= V_L I_L [2 \cos 30 \cos \phi] \\ &= V_L I_L \sqrt{3} \cos \phi \\ P &= \sqrt{3} V_L I_L \cos \phi \quad (4) \end{aligned}$$

Power factor calculation

We know that,

$$W_1 = V_L I_L \cos(30 + \phi)$$

$$W_2 = V_L I_L \cos(30 - \phi)$$

$$\text{power} = W_1 + W_2$$

$$= \sqrt{3} V_L I_L \cos \phi$$

Find

$$\begin{aligned} W_2 - W_1 &= V_L I_L \cos(30 - \phi) - V_L I_L \cos(30 + \phi) \\ &= V_L I_L [\cos(30 - \phi) - \cos(30 + \phi)] \\ &= V_L I_L [\cos 30 \cdot \cos \phi + \sin 30 \cdot \sin \phi - \cos 30 \cdot \cos \phi + \sin 30 \cdot \sin \phi] \\ &= V_L I_L [2 \sin 30 \cdot \sin \phi] \\ &= V_L I_L [2 \frac{1}{2} \sin \phi] \end{aligned}$$

$$W_2 - W_1 = V_L I_L \sin \phi$$

$$\frac{W_2 - W_1}{W_1 + W_2} = \frac{V_L I_L \sin \phi}{\sqrt{3} V_L I_L \cos \phi}$$

$$= \frac{1}{\sqrt{3}} \tan \phi$$

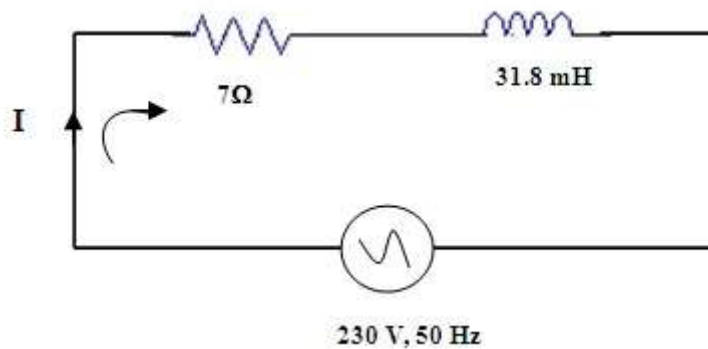
$$\text{i.e. } \tan \phi = \sqrt{3} \left(\frac{W_2 - W_1}{W_2 + W_1} \right)$$

$$\text{phase angle } \phi = \tan^{-1} \left[\sqrt{3} \frac{(W_2 - W_1)}{(W_2 + W_1)} \right]$$

$$\text{power factor } \cos \phi = \cos \left[\tan^{-1} \left(\sqrt{3} \frac{(W_2 - W_1)}{(W_2 + W_1)} \right) \right]$$

Part-B Jan 2013

2.19 A coil having a resistance of 7Ω and inductance of 31.8 mH is connected to 230 V , 50 Hz supply. Calculate the circuit current power factor and power consumed.



Current

$$X_L = \omega L = 2\pi f \times L$$

$$= 2 \times \pi \times 50 \times 31.8 \times 10^{-3}$$

$$= 9.99\Omega$$

$$Z = R + jX_L = 7 + j9.99 = 12.19 \angle 54.98^\circ$$

$$I = \text{current} = \frac{V}{|Z|}$$

$$= \frac{230}{12.19}$$

$$= 18.86 \text{ A}$$

Power factor $\cos \phi$

$$\phi = \tan^{-1} \frac{X_L}{R}$$

$$= \tan^{-1} \frac{9.99}{7} = 54.98^\circ$$

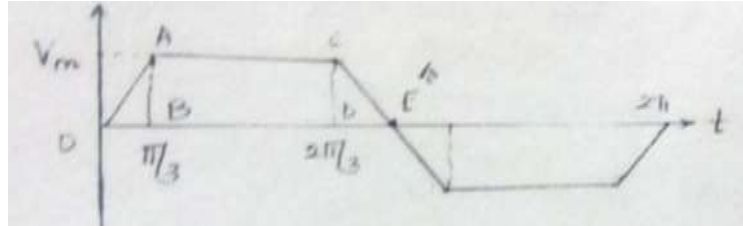
$$\cos \phi = \cos 54.98^\circ = 0.573$$

Power consumed

$$\begin{aligned}
 P &= VI \cos \theta \\
 &= 230 \times 18.86 \times 0.573 \\
 P &= 2489.23W \\
 P &\approx 2.4kW
 \end{aligned}$$

Part – B (April 2010)

2.20 Find the average, RMS, form factor and peak factor of the following waveform.



$$\begin{aligned}
 V_{avg} \text{ of } \Delta_1 &= \frac{\frac{1}{2}bh}{\text{base}} \\
 \Delta \rightarrow h &= V_m; b = \pi/3
 \end{aligned}$$

Note

$$\frac{2\pi}{3} - \frac{\pi}{3} = \frac{2\pi - \pi}{3} = \frac{\pi}{3}$$

$$\begin{aligned}
 V_{avg} \text{ of Triangle 1} &= \frac{\frac{1}{2} \times \pi/3 \times V_m}{\pi/3} \\
 &= V_m/3
 \end{aligned}$$

$$\begin{aligned}
 V_{avg} \text{ of Rectangle} &= \frac{l \times b}{\text{base}} \\
 &= \frac{V_m \times \frac{\pi}{3}}{\frac{\pi}{3}} - V_m
 \end{aligned}$$

Note

$$\pi - \frac{2\pi}{3} = \frac{3\pi - 2\pi}{3} = \frac{\pi}{3}$$

$$\begin{aligned}
 V_{avg} \text{ of Triangle 2} &= \frac{\frac{1}{2} \times b \times h}{\text{base}} \\
 &= \frac{\frac{1}{2} \times \pi/3 \times V_m}{\pi/3} \\
 &= V_m/2
 \end{aligned}$$

$$\begin{aligned}
 V_{avg} &= V_{avg_1} + V_{avg_2} + V_{avg_3} \\
 &= \frac{V_m}{2} + V_m + \frac{V_m}{2} \\
 &= 2V_m.
 \end{aligned}$$

$$\begin{aligned}
 \text{Average value} &= \frac{V_{avg}}{\text{no. of shape}} \\
 &= \frac{2V_m}{3}
 \end{aligned}$$

$$V_{RMS} = \sqrt{\frac{1}{\pi} \left(\frac{1}{3}bh^2 + l^2b + \frac{1}{3}bh^2 \right)}$$

RMS Value

$$\begin{aligned}
 &= \sqrt{\frac{1}{\pi} \left(\frac{1}{3} \times \frac{\pi}{3} \times V_m^2 + V_m^2 \times \frac{\pi}{3} + \frac{1}{3} \times \frac{\pi}{3} \times V_m^2 \right)} \\
 &= \sqrt{\frac{1}{\pi} \left(\frac{\pi}{9} V_m^2 + V_m^2 \frac{\pi}{3} + \frac{\pi}{9} V_m^2 \right)} \\
 &= \sqrt{\frac{1}{\pi} \times \frac{\pi}{3} V_m^2 \left(\frac{1}{3} + 1 + \frac{1}{3} \right)} \\
 &= \sqrt{\frac{V_m^2}{3} \left(\frac{5}{3} \right)} \\
 &= \sqrt{\frac{5}{9} V_m^2} = \frac{\sqrt{5}}{3} V_m
 \end{aligned}$$

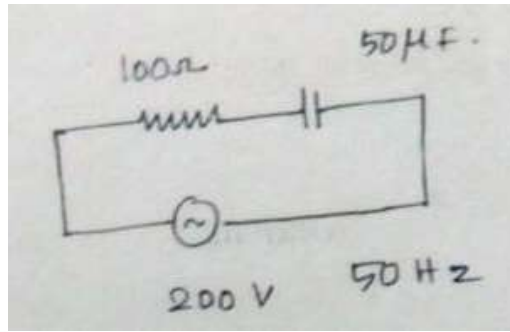
Form factor

$$\begin{aligned}
 &= \frac{\text{RMS value}}{\text{average value}} \\
 &= \frac{\sqrt{\frac{5}{3}} V_m}{\frac{2}{3} V_m} \\
 &= \frac{2}{\sqrt{5}} = 1.34
 \end{aligned}$$

Peak factor

$$\begin{aligned}
 &= \frac{V_m}{\sqrt{\frac{5}{3}} V_m} \\
 &= \frac{2}{\sqrt{5}} = 1.34
 \end{aligned}$$

2.21 A resistance of 100Ω is connected in series with a $50\mu\text{F}$ capacitor to a supply at 200V, 50Hz. Find the a) impedance, current, power factor and phase angle, b) voltage across resistor and capacitor, c) phasor diagram

Solution**Impedance**

$$Z = R - jX_C$$

$$X_C = \frac{1}{\omega_C}$$

$$= \frac{1}{2\pi fX_C}$$

$$= \frac{1}{2\pi \times 50 \times 50 \times 10^{-6}}$$

$$X_C = 63.66\Omega$$

$$Z = 100 - j63.66 = 118.5 \angle -32.4^\circ$$

$$\text{current} = \frac{V}{Z} = \frac{200}{118.5} = 1.68\text{A}$$

Phase angle

$$\phi = \tan^{-1} \frac{X_C}{R}$$

$$= \tan^{-1} \frac{63.66}{100}$$

$$\phi = 32.48^\circ$$

Power factor

$$= \cos \phi$$

$$= \cos 32.48 = 0.83$$

Voltage across resistor and capacitor**Voltage across resistor**

$$V_1 = I \times R$$

$$= 1.68 \times 100$$

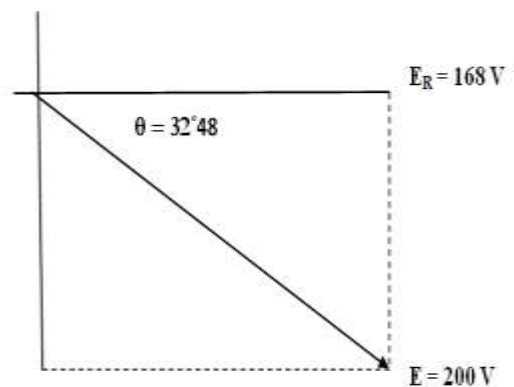
$$V_1 = 168\text{V}$$

Voltage across capacitor

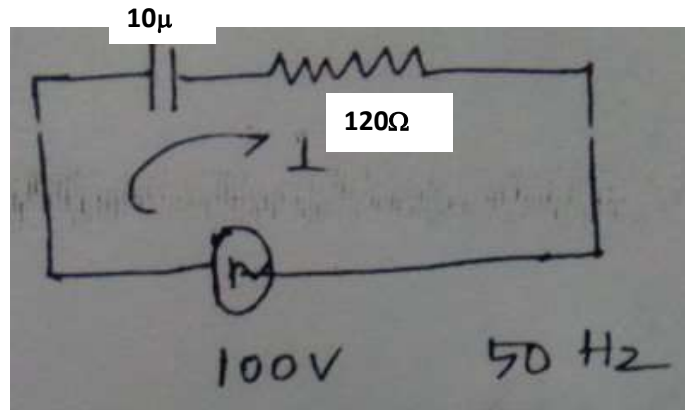
$$V_2 = I \times X_C$$

$$= 1.68 \times 63.66$$

$$V_2 = 106.94\text{V}$$

**Phasor Diagram**

2.22 A capacitor having capacitance of $10\mu\text{F}$ is connected in series with a pure resistance of 120Ω across 100V , 50 Hz supply, calculate current, phase angle ϕ , power consumed



Current

to find the net impedance

$$Z = R - jX_C$$

$$X_C = \frac{1}{\omega_C}$$

$$= \frac{1}{2\pi f X_C}$$

$$= \frac{1}{2\pi \times 50 \times 10 \times 10^{-6}}$$

$$X_C = 318.30\Omega$$

$$Z = 120 - j318.30 = 340.16 \angle -69.3\Omega$$

$$\text{current} = \frac{V}{Z} = \frac{100}{340.16} = 0.29\text{A}$$

Phase angle

$$\phi = \tan^{-1} \frac{X_C}{R}$$

$$= \tan^{-1} \frac{318.3}{120}$$

$$\phi = 69.34^\circ$$

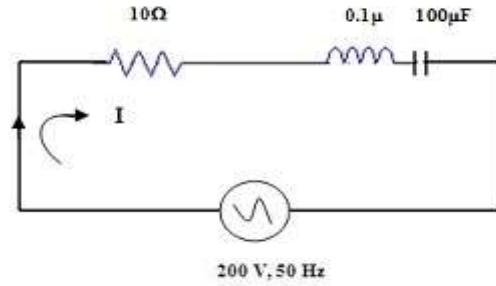
Power consumed

$$P = VI \cos \phi$$

$$= 100 \times 0.29 \times \cos 69.34 = 10.23\text{W}$$

2.23 A coil of resistance 10Ω and inductance if 0.1H is connected in a series with a condenser of capacitance $150\mu\text{F}$ across 200 V , 50Hz supply calculate

- a) impedance
- b) current
- c) power factor

**Impedance**

$$Z = R + j(X_L - X_C)$$

$$X_L = \omega L$$

$$= 2\pi f X_L$$

$$= 2 \times \pi \times 50 \times 0.1$$

$$X_L = 31.41\Omega$$

$$X_C = \frac{1}{\omega C}$$

$$= \frac{1}{2\pi f X_C}$$

$$= \frac{1}{2\pi \times 50 \times 150 \times 10^{-6}}$$

$$X_C = 21.22\Omega$$

$$Z = 120 + j(31.41 - 21.22) = 10 + j10.19 = 14.27 \angle 45$$

$$\text{current} = \frac{V}{Z} = \frac{200}{14.27} = 14.01\text{A}$$

Power factor

$$\phi = \tan^{-1} \left(\frac{X_L - X_C}{R} \right)$$

$$= \tan^{-1} \left(\frac{31.41 - 21.22}{10} \right)$$

$$\phi = 45.53^\circ$$

$$\begin{aligned} \text{Power factor } \cos \phi &= \cos 45.53 \\ &= 0.700 \end{aligned}$$

Part A (Nov 2011)**Two Marks**

1. write the relationship between line and phase value of Y.

Y – Star connection system.

The line current = phase current

$$I_L = I_{ph}$$

The line voltage $\neq$ phase voltage

$$\text{i.e., } V_{line} = \sqrt{3}V_{phase}$$

2. Write the demerits of three wattmeter method.

- It requires three wattmeter to measure 3 ϕ power
- Three wattmeter connection to the three phase system is difficult to make because usually delta may not be broken and also the star point may not be available.

3. Write short notes on wattmeter

Wattmeter is an electrical device which is used to sense voltage, current and phase angle between voltage and current and shows the power in watts.

a. Current coil:

- always connected in series with load
- resistance is low

b. Pressure coil:

- senses voltage and connected across the supply.
- Resistance is high.

4. Advantages and disadvantages of 2 wattmeter method

Advantages

- It is applicable for balanced and unbalanced 3 ϕ load
- only 2 wattmeter are needed to measure 3 ϕ power
- not only power, but power factor can be determined
- Neutral power point for star connected load is not necessary to connect with wattmeter.
- Delta connected load need not be opened for connecting to Watt meter

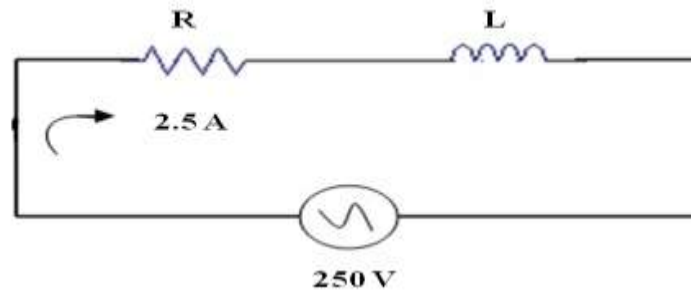
Disadvantages

- Not applicable for 3 phase, 4 wire system
- W_1 and W_2 sign must be identified correctly or it will lead to wrong result.

(Part B Nov 2011)

2.15 A choke coil takes a current of 2.5A when connected across 250V main and consumes 400 W. Find

- a) power factor,
- b) resistance of the coil,
- c) inductance of coil
- d) apparent power,
- e) reactive power.



Voltage across resistance $V_R = IR$

Voltage across inductance $V_L = IX_L$

$$V = V_R + V_L$$

$$V = IR + jIX_L$$

$$V = I(R + j2\pi \times fL)$$

$$250 = 2.5(R + j2\pi \times fL)$$

$$Z = (R + j2\pi fL) = \frac{250}{2.5} = 100\Omega$$

$$Z = \sqrt{R^2 + X_L^2}$$

$$Z = \theta = \tan^{-1}\left(\frac{X_L}{R}\right)$$

$$P : \text{power} = VI \cos \theta$$

$$400(\text{real power}) = 250 \times 2.5 \cos \theta$$

$$\cos \theta = \frac{400}{(250 \times 2.5)}$$

$$\cos \theta = 0.64$$

$$\theta = \cos^{-1} 0.64 = 50.20$$

$$\theta = \tan^{-1}\left(\frac{X_L}{R}\right)$$

$$50.20 = \tan^{-1}\left(\frac{X_L}{R}\right)$$

$$\tan^{-1}(50.20) = \frac{X_L}{R}$$

$$1.20 = \frac{X_L}{R}$$

$$X_L = 1.20R$$

$$2\pi fX_L = 1.20$$

$$X_L = \frac{1.20}{2 \times \pi \times f}$$

$$X_L = 3.8 \times 10^{-3} \times R$$

$$X_L = 1.20R$$

$$Z = \sqrt{R^2 + (1.20R)^2}$$
$$= \sqrt{R^2 + (1.20R)^2}$$

$$100 = R\sqrt{2.44}$$

$$100 = R(1.56)$$

$$R = \frac{100}{1.56}$$

$$R = 64.01\Omega$$

$$X_L = 1.20 \times R$$

$$= 1.2 \times 64.01$$

$$= 76.82$$

$$2\pi fL = 76.82$$

$$L = \frac{76.82}{2 \times \pi \times 50}$$

$$= 0.244 = 24mH$$

$$L = 24mH$$

Reactive Power

$$Q = VI$$

$$= 250 \times 2.5 = 625VA$$

Apparent Power

$$= |P + jQ|$$

$$= |400 + j625|$$

$$= 742.04VA$$